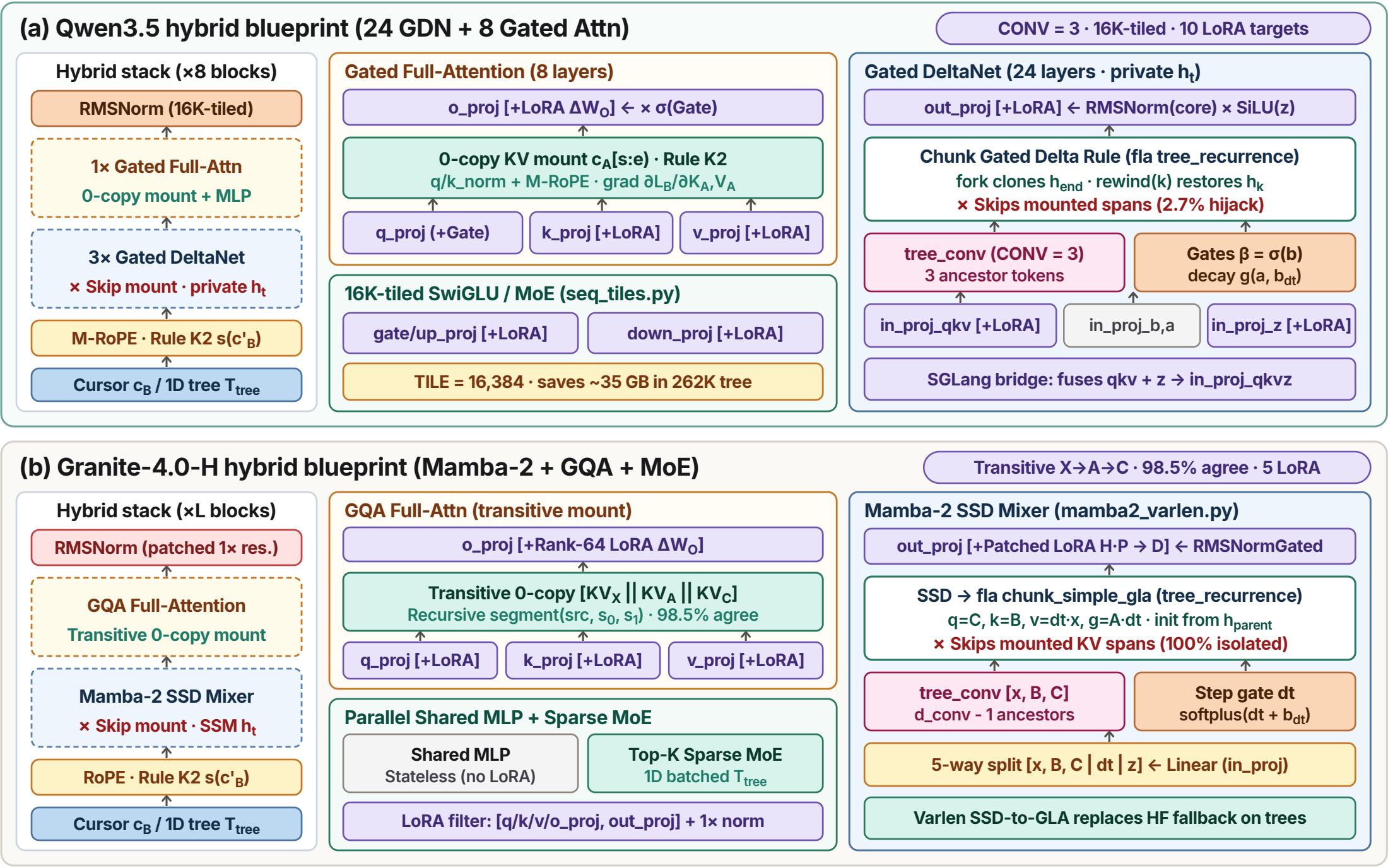
(a) Qwen3.5 hybrid blueprint (24 GDN + 8 Gated Attn)
CONV = 3 · 16K-tiled · 10 LoRA targets
Hybrid stack (×8 blocks)
RMSNorm (16K-tiled)
1× Gated Full-Attn
0-copy mount + MLP
3× Gated DeltaNet
× Skip mount · private h_t
M-RoPE · Rule K2 s(c'_B)
Cursor c_B / 1D tree T_tree
Gated Full-Attention (8 layers)
o_proj [+LoRA ΔW_O] ← × σ(Gate)
0-copy KV mount c_A[s:e) · Rule K2
q/k_norm + M-RoPE · grad ∂L_B/∂K_A,V_A
q_proj (+Gate)
k_proj [+LoRA]
v_proj [+LoRA]
16K-tiled SwiGLU / MoE (seq_tiles.py)
gate/up_proj [+LoRA]
down_proj [+LoRA]
TILE = 16,384 · saves ~35 GB in 262K tree
Gated DeltaNet (24 layers · private h_t)
out_proj [+LoRA] ← RMSNorm(core) × SiLU(z)
Chunk Gated Delta Rule (fla tree_recurrence)
fork clones h_end · rewind(k) restores h_k
× Skips mounted spans (2.7% hijack)
tree_conv (CONV = 3)
3 ancestor tokens
Gates β = σ(b)
decay g(a, b_dt)
in_proj_qkv [+LoRA]
in_proj_b,a
in_proj_z [+LoRA]
SGLang bridge: fuses qkv + z → in_proj_qkvz
(b) Granite-4.0-H hybrid blueprint (Mamba-2 + GQA + MoE)
Transitive X→A→C · 98.5% agree · 5 LoRA
Hybrid stack (×L blocks)
RMSNorm (patched 1× res.)
GQA Full-Attention
Transitive 0-copy mount
Mamba-2 SSD Mixer
× Skip mount · SSM h_t
RoPE · Rule K2 s(c'_B)
Cursor c_B / 1D tree T_tree
GQA Full-Attn (transitive mount)
o_proj [+Rank-64 LoRA ΔW_O]
Transitive 0-copy [KV_X || KV_A || KV_C]
Recursive segment(src, s_0, s_1) · 98.5% agree
q_proj [+LoRA]
k_proj [+LoRA]
v_proj [+LoRA]
Parallel Shared MLP + Sparse MoE
Shared MLP
Stateless (no LoRA)
Top-K Sparse MoE
1D batched T_tree
LoRA filter: [q/k/v/o_proj, out_proj] + 1× norm
Mamba-2 SSD Mixer (mamba2_varlen.py)
out_proj [+Patched LoRA H·P → D] ← RMSNormGated
SSD → fla chunk_simple_gla (tree_recurrence)
q=C, k=B, v=dt·x, g=A·dt · init from h_parent
× Skips mounted KV spans (100% isolated)
tree_conv [x, B, C]
d_conv - 1 ancestors
Step gate dt
softplus(dt + b_dt)
5-way split [x, B, C | dt | z] ← Linear (in_proj)
Varlen SSD-to-GLA replaces HF fallback on trees